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# The effect of abstract inter-chunk relationships on serial-order control

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ABSTRACT

Hierarchical control is often thought to dissect a complex task space into isolated subspaces in order to eliminate interference. Yet, there is also evidence from serial-order control tasks that our cognitive system can make use of abstract relationships between different parts (chunks) of a sequence. Past evidence in this regard was limited to situations with ordered stimuli (e.g., numbers or positions) that may have aided the detection of relationships and allowed gradual learning and hypothesis testing. Therefore, we used a modified task-span paradigm (with no ordered elements between tasks) in which participants performed memorized sequences of tasks that were encoded in terms of separate chunks of three tasks each. To allow examination of learning effects, each sequence was "cycled" through repeatedly. Importantly, we compared sequences whose chunks were governed by a common, abstract grammar with sequences whose chunks were governed by different grammars. Experiment 1 examined the effect of relationships between shared-element chunks (e.g., ABB-BAA vs. ABB-BAB), Experiment 2 and 3 between different-element chunks (e.g., ABA-CDC vs. ABA-CCD), and Experiment 4 examined second-order relationships (e.g., ABA-ABB—CDC-CDD vs. ABA-ABB—CDC-CCD). Robust evidence in favor of beneficial effects of abstract inter-chunk relationships was obtained across all four experiments. Importantly, these effects were at least as strong in initial cycles of performing a given sequence as during later cycles, suggesting that the cognitive system operates with an "expectation of abstract relationships," rather than benefitting from them through gradual learning. We discuss the implications of these results for models of hierarchical control.

## 1. Introduction

Since the beginning of the cognitive revolution, hierarchical control has been viewed as a core element of human cognition that enables complex action patterns. Recently, this interest has been revived, mainly through neuroscience evidence suggesting a neuroanatomical reality to hierarchical levels. Theories of hierarchical control that have developed alongside the neural evidence usually suggest that higher levels of control represent the context or rules that shape the current, lower-level processes. On each level, selection of the currently relevant representation is shaped by the context provided by the representations in the next-higher level. The elements on a given level are typically treated as individual entities that are selected one at a time, often assuming a winner-take-all selection mechanism and therefore to the exclusion of competing entities on the same level (Badre, 2008; Duncan, 2010; Koechlin, Ody, & Kouneiher, 2003; Miller, Galanter, & Pribram, 1960; Rosenbaum, Kenny, & Derr, 1983). This strategy provides a "divide-and-conquer" approach to complex task spaces: By carving a larger task space into separate, manageable sub-spaces, interference can be eliminated and lower-level sub-spaces can be flexibly re-combined to solve new problems (Badre & Nee, 2018).

Notions of hierarchical control have been particularly important in the context of serial-order control of action sequences, such as playing the piano, typing, or processing language. People tend to proceed through complex sequences in "chunks" of 3–4 basic elements (e.g., notes in a piece of music), rather than in an element-by-element manner (Lashley, 1951). Chunks, in turn, can be organized into larger sub-sequence plans, and so forth (Dehaene, Meyniel, Wacongne, Wang, & Pallier, 2015; Krampe, Mayr, & Kliegl, 2005; Lashley, 1951). The borders of chunk representations within a larger sequence can be identified by higher RTs and lower accuracy rates at the chunk transition positions (Lien & Ruthruff, 2004; Schneider & Logan, 2006; Wu, He, Zhou, Xiao, & Luo, 2017).

From the perspective of models that emphasize the "divide-and-conquer" view of hierarchical control, these types of behavioral patterns can be explained by assuming that on each level only one chunk or plan is active at a time, specifying the sequence of within-chunk elements on the level below. This comes with the implicit assumption that a chunk-

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level code "knows nothing" about its contents, that is the specific elements it contains, or of the serial-order or content in competing chunks (Fitch & Martins, 2014). Such knowledge transfer between chunks or levels would create exactly the kind of interference across regions of the task space that the hierarchical "divide and conquer" strategy is supposed to eliminate.

Interestingly however, much of the early literature on serial-order control indicates that abstract relationships between the content of chunks on the same level impact behavior. Such evidence comes from studies requiring participants to learn sequences of spatial locations, which either followed abstract patterns or not, from feedback. Most notably, Restle (1970) showed that people use relationships across consecutive parts of a sequence, such that performance with transposed elements (e.g., 1–2-3—4-5-6, where numbers stand for left-to-right positions) was better than with sequences without such abstract, chunk-to-chunk relationships (see also, Collard & Povel, 1982; Koch & Hoffmann, 2000; Povel & Collard, 1982). Based on these, and similar results, Dehaene et al. (2015) argued that chunks are encoded in a relational manner, that is terms of their abstract grammatical schemas, or "algebraic patterns defined by identity relationships," (p. 9) and then organized within nested (hierarchical) tree structures.

The notion that the cognitive system represents sequences in general by coding them in terms of abstract, inter-chunk relationships would produce theoretical tension with models that focus mainly on the need for negotiating interference and mutual competition between elements on the same hierarchical level. Yet, effects of inter-chunk relationships on performance have only been demonstrated using learning paradigms in which participants were asked to "discover" a given sequence through hypothesis testing and feedback. Further, this previous work always used stimuli with an inherent order, such as sequences of spatial locations. With these limitations, it is not clear whether the appreciation of abstract inter-chunk patterns is a general feature of serial-order representations, or if it is limited to learning situations and ordered sets of stimuli.

The presence of natural ordering principles among sequential elements implies that within individual chunks, relational patterns between elements can be established, upon which inter-chunk relationship can be built. In the absence such ordering principles, the only remaining feature for establishing abstract patterns is that of repetitions and alternations between sequential elements within and between chunks. Therefore, we used as sequential elements pairs of S-R rules that have no inherent ordering. Within each of our three-rule chunks, there was at least one rule repetition (e.g., AAB, ABB or ABA, where letters A and B stand for a possible pair of rules) that defined the chunk grammar. Two chunks that make up a 6-element sequence could either be defined by the same (matching) grammars (e.g., AAB-BBA), or different (non-matching) grammars (e.g., AAB-BAB).

Following Restle (1970) and more recently, Dehaene, Al Roumi, Lakretz, Planton, and Sablé-Meyer (2022), from the relational-coding perspective one can think of each chunk as a mental program for how to perform its individual elements by specifying the relations between them. For example, the program for chunk ABB might be "start with A, then change, then repeat." Further, if the grammar of the next chunk is the same, but with a different starting element (or different element pairs altogether), then the same set of instructions can be re-used, with a new starting element (e.g., the sequence BAA would be "start with B, then change, then repeat"). Assuming that chunks are represented in this manner, sequences with chunks that can use the same program should be easier to perform than sequences whose chunks require different programs.

Our first and most general question then is to what degree sequential performance benefits from matching chunk grammars in the absence of ordered sets of elements. A confirmatory result would indicate that our cognitive system tries to exploit any across-chunk relationships in order to maximize efficiency of sequential representations (Dehaene et al., 2015).

A second question we will try to answer is about the points in a sequence at which matching grammars influence performance. One possibility suggested by previous evidence (e.g., Restle, 1970) is that benefits occur predominantly during transitions between chunks. Such a pattern would indicate that the matching instructions can be used to plan entire upcoming chunks, thereby eliminating or reducing the need for selecting within-chunk, sequential elements. However, as we will elaborate in the introduction to Experiment 2, the evidence for strong chunk-transition benefits comes exclusively from studies using elements that were not only inherently ordered, but were also shared across chunks. In such cases, one chunk can be transformed into the next in a very straightforward manner. It is therefore important to examine cases in which chunks are governed by similar abstract grammars, but where the lack of shared elements does not allow such direct transformations.

Our final and theoretically most important question is to what degree an appreciation of abstract patterns emerges over time, or whether such patterns can be exploited from the outset. Given that previous work in this area has used learning paradigms in which relationships between chunks could be discovered either implicitly (e.g., Koch & Hoffmann, 2000), or through feedback (Restle, 1970), the question of gradual learning versus instantaneous use of abstract chunk relationships has yet to be addressed empirically. Therefore, we presented to participants in each block a new sequence of rules to be cycled through repeatedly for the duration of the block, allowing us to track the trajectory of matching-grammar effects across sequence repetitions.

A result showing gradual learning would be consistent with influences of implicit or procedural learning on executive control (Marcus, Vijayan, Bandi Rao, & Vishton, 1999; Oberauer, Souza, Druey, & Gade, 2013; Trach, McKim, & Desrochers, 2021). There is also evidence from both neurologically plausible network modeling and behavioral experiments that abstract hierarchical structures can be inferred through gradual learning (Collins & Frank, 2013). In contrast, results showing instantaneous benefits from matching grammars would indicate that the cognitive system has an a-priori "preparedness" for coding sequential elements in a relational manner (Dehaene et al., 2022). In this case, relations between elements and between chunks could be detected during the initial encoding of sequential elements and then used to establish effective retrieval structures for subsequent sequence production (Ericsson & Kintsch, 1995).

## 2. Experiment 1: Shared-element chunks

As mentioned, previous results that indicated relational coding of chunks have come from experimental paradigms in which within-chunk elements had an inherent order and the same elements could occur across chunks (e.g., Restle, 1970). In this first experiment we wanted to take a step towards generalizing the relational coding finding to sequences without ordered elements. However, the same elements were still used across chunks. Specifically, we used sequences constructed from two three-element chunks of stimulus-response (S-R) rules, with no inherent order between different S-R rules.

Thus, we constructed sequences that followed relatively closely the types of sequences used in previous work, in that in our "matching" sequences, abstract relationships were used to translate one part of the sequence (i.e., chunk) into the next. However, given that we used no ordered set of stimuli, no arithmetic transformations could be applied here.

### 2.1. Methods

#### 2.1.1. Participants

Data were collected from 56 undergraduate students at the University of Oregon. Existing research provides no strong guidance for effect-size estimates. Therefore, we chose a sample size that provided sufficient power (0.8) for detecting small to moderate effect sizes ($f$ = 0.15). Participation was voluntary, and students were offered extra credit for their participation.

#### 2.1.2. Spatial rules task

We applied an explicit sequencing paradigm to a simple spatial rules task in order to test our question about the role of matching chunk patterns in sequential performance (Mayr, 2002, 2009). On every trial of the spatial rules task, a white dot would appear in one corner of a frame, and participants were instructed to apply a spatial rule to the dot (e.g., horizontal) to determine which corner the dot would move to. Responses were made using the 4, 5, 1, and 2 on the keyboard number pad, with each key representing the corresponding corner of the frame (top left, top right, bottom left, bottom right, respectively). In this task, all task features (stimulus, rule, etc.) are orthogonal to one another.

#### 2.1.3. Sequencing paradigm and procedure

Sequences were constructed from the two different spatial rules as elements. These rule sequences allow more precise temporal control over participants' sequential performance (as opposed to using response sequences). Further, response sequences would utilize inherently ordered information, whereas different spatial rules have no obvious order and therefore allow us to establish truly abstract chunk grammars. Each rule sequence contained two 3-element chunks. For each new sequence, participants were presented with the sequence of rules listed vertically in an instruction screen (see Fig. 1). In order to visually enforce the chunk structure within each sequence, there was a small space between the first three rules (chunk 1) and the second three rules (chunk 2). Past work has shown that such visual cuing is highly effective in inducing chunking patterns (e.g., Anderson & Matessa, 1997; Mayr, 2009; Schneider & Logan, 2006). See Appendix A for a list of all possible sequences. Participants could study this instruction screen as long as they wanted before pressing the space bar to initiate the first trial of the block. Participants cycled through each sequence, applying one rule per trial, 8 times, for 18 different six-element sequences (48 total trials per sequence). After each response there was a 100 ms inter-trial interval before the next stimulus was displayed. Following incorrect responses, the sequence of rules would be displayed above the frame, with the incorrect response colored red. Participants were then required to correct their mistake before moving on to the next trial. Note that despite the error feedback, this paradigm was not a feedback-driven learning task. Rather, the feedback ensured that participants could "get back into the sequence" and produce analyzable data after errors (which were overall quite rare).

Chunk patterns, or grammars, were defined by the order of their elements (rules), with matching chunks defined as those with the same abstract chunk grammar, regardless of what specific elements were in each position in the chunk. For this first, more conservative definition of grammars, the two chunks in a sequence contained the same pair of rule elements, horizontal and vertical (e.g., ABB-BAA). One-third of the sequences were "matching" and the other two-thirds were "non-matching." Participants completed this computerized task in a one-hour session.

### 2.2. Results

All data and analysis scripts relevant for this project are provided at https://osf.io/xuy6a/?view_only=6ecf656a428748d5a9c940e1f0473579.

The first cycle of each new sequence was removed as practice (i.e., the first six trials of each block), as well as trials with RTs below 100 ms or above the 99.5th percentile, and those following error trials. Further, the data for participants who did not complete the study or had accuracy rates below 70% were not used in analysis. Two participants did not complete the study, and one participant did not meet the minimum accuracy inclusion criterion, and so the data from 53 participants were used in analyses.

Fig. 2 (first column) shows each of the three performance outcomes as a function of sequence position and relationship between chunks (matching vs. non-matching grammars). Consistent with the notion that sequential performance is governed by hierarchically organized representations, the RT pattern shows strong and very typical sequence and chunk transition effects (see also, Mayr, 2009; Schneider & Logan, 2006). It is also apparent in the figure that RTs were faster and error rates smaller for sequences with matching versus non-matching chunks, and for RTs, that effect appears strongest at chunk transition points (first and fourth position in the sequence). We ran linear mixed-effects models with random slopes for grammar matching and random intercepts for subjects (Bates, Mächler, Bolker, & Walker, 2015; R Core Team, 2018). Here and in all remaining experiments, we also confirmed that the pattern of results using standard analyses of variance with subject/condition averaged dependent variables.

Though correct trial RTs (log-transformed) and accuracy rates were the primary dependent variables, potential speed-accuracy tradeoffs could introduce unnecessary ambiguity into our findings. Therefore, we created a standardized performance cost variable by first z-scoring error rates and log-transformed RTs across participants and conditions, and then averaging the two z-scores into a composite performance cost measure (see Liesefeld & Janczyk, 2019). For this performance cost variable, as is the case for both RTs and error rates, higher numbers represent worse performance. Each of the three dependent variables was predicted by element position variables, abstract grammar match, and their interaction as fixed effects. The element position variables aimed to capture potential transition effects on different levels of the hierarchical sequential representation using the following contrasts applied across sequence positions: The chunk transition contrast (CT) compared positions 1 and 4 against positions 2, 3, 5, and 6. The within-chunk contrast (WC) tested positions 2 and 5 against positions 3 and 6). Finally, the sequence transition contrast (ST) compared the chunk transition of the first chunk (position 1) against the chunk transition of the second chunk (position 4) to test the effect of highest-level transition within these sequences. We are most interested here in the CT and the ST contrasts, which test for the effects of chunk-level and sequence-level transitions.

The results of these analyses are presented in Table 1. As is apparent from the significant main effect of matching grammars on all three outcomes, they confirm that matching chunk grammars indeed lead to

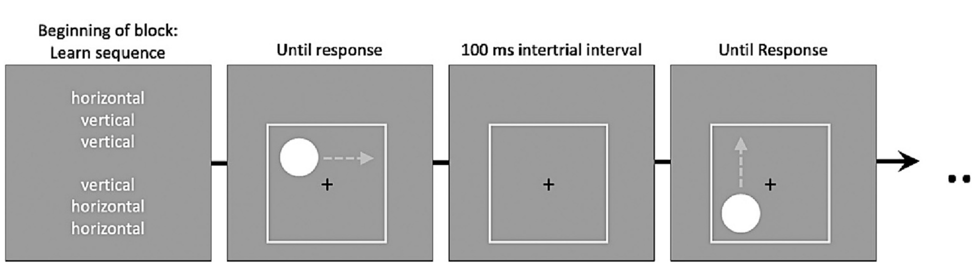


**Fig. 1.** Sequence of events in the task-span procedure in Experiment 1, with explicitly instructed spatial rules. Each sequence of six rules was presented at the beginning of a block. After pressing any key, participants would begin cycling through the sequence, applying one rule per trial of the spatial rules task. Gray dashed arrows indicate correct responses and were not presented in the actual trials. In this example, the sequence of rules contains matching chunk grammars.

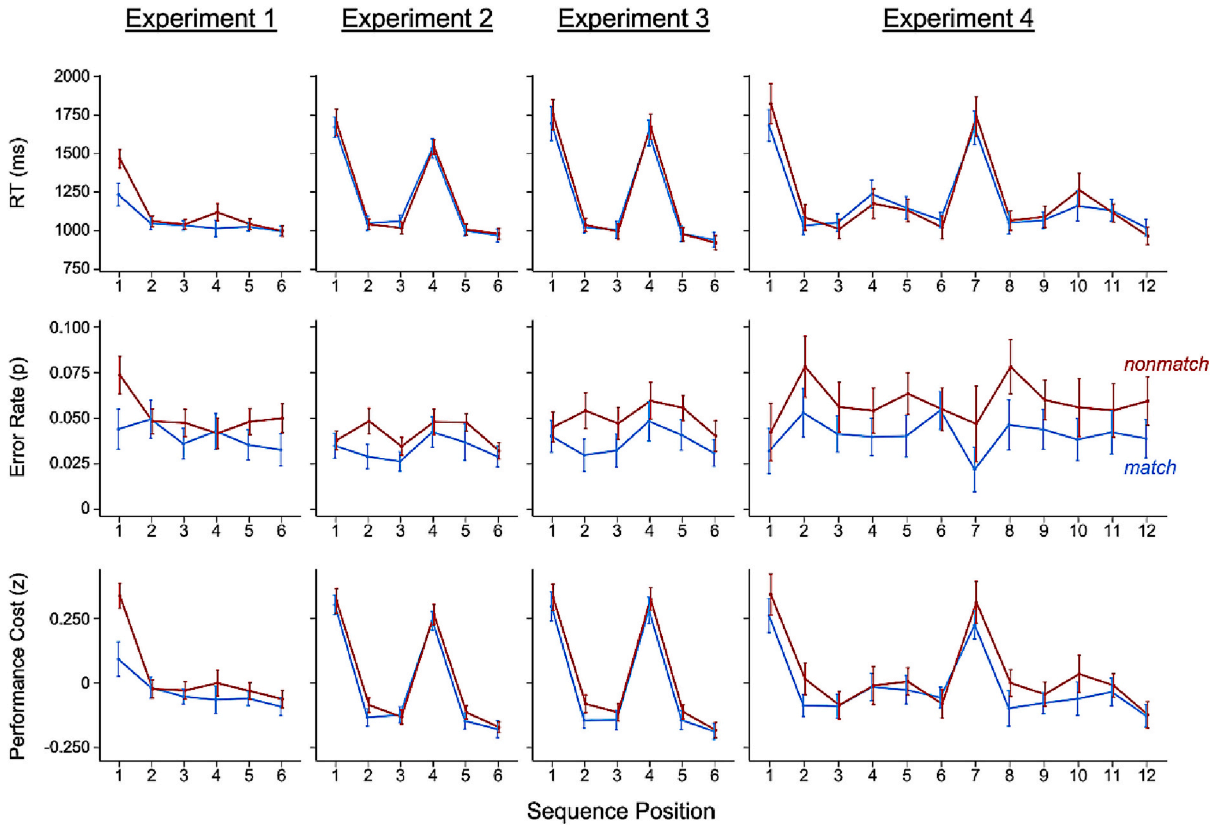


**Fig. 2.** RTs (top panels), error rates (middle panels), and standardized performance costs (bottom panels) for each experiment, at all positions in matching versus non-matching sequences. Error bars represent 95% within-subject confidence intervals.

**Table 1**
Fixed effects from linear mixed models for Experiment 1.

| | RT | | | | Error Rate | | | | Standardized Performance Score | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | B | SE | t | p | B | SE | z | p | B | SE | t | p |
| Match | −0.060 | 0.015 | −3.90 | <0.001 | −0.284 | 0.076 | −3.75 | <0.001 | −0.087 | 0.020 | −4.33 | <0.001 |
| CT | 0.094 | 0.003 | 29.34 | <0.001 | 0.073 | 0.031 | 2.38 | 0.018 | 0.102 | 0.009 | 11.70 | <0.001 |
| WC | 0.010 | 0.004 | 2.63 | 0.009 | −0.003 | 0.036 | −0.08 | 0.932 | 0.009 | 0.010 | 0.91 | 0.362 |
| ST | 0.135 | 0.005 | 25.79 | <0.001 | 0.302 | 0.049 | 6.15 | <0.001 | 0.169 | 0.014 | 11.83 | <0.001 |
| Match * CT | −0.066 | 0.005 | −11.96 | <0.001 | −0.010 | 0.056 | −0.17 | 0.863 | −0.068 | 0.012 | −5.48 | <0.001 |
| Match * WC | −0.002 | 0.006 | −0.31 | 0.756 | 0.116 | 0.068 | 1.71 | 0.086 | 0.007 | 0.014 | 0.52 | 0.601 |
| Match * ST | −0.058 | 0.009 | −6.47 | <0.001 | −0.289 | 0.090 | −3.21 | 0.001 | −0.090 | 0.020 | −4.47 | <0.001 |
| Model Stats | AIC | $cR^2$ | $mR^2$ | ICC | AIC | $cR^2$ | $mR^2$ | ICC | AIC | $cR^2$ | $mR^2$ | ICC |
| | 47,535 | . 189 | 0.039 | 0.157 | 13,924 | 0.100 | 0.012 | 0.089 | −341 | 0.725 | 0.155 | 0.674 |

*Note.* For each of the outcomes, the following model was used: Outcome ~ Match * Position Contrasts + (1 + Match|Subject). Match = matching grammars within sequence. CT = chunk transition contrast: first and fourth elements vs. all others (second, third, fifth, and sixth). WC = within-chunk contrast: second and fifth elements vs. third and sixth. ST = sequence transition contrast: first vs. fourth element. AIC = Akaike information criterion, $cR^2$ = conditional $R^2$, $mR^2$ = marginal $R^2$, ICC = intraclass correlation coefficient.

performance benefits. For RTs and standardized performance scores, matching effects were strongest at transition points. For errors, there were specific effects for the highest-level sequence transition, in addition to the overall, position-unspecific matching effect.

As a second step, we examined whether the grammar matching effect develops gradually through experience with a given sequence, or instead is present from the outset. For this purpose, we added within-block sequence repetitions as an additional predictor in the form of linear and quadratic polynomial contrasts that were allowed to interact with the chunk match variable. No statistically significant effects were obtained. In particular, the interactions between the repetition contrasts and the chunk match factor were not reliable for RTs (linear $t = -0.21$, $p = .834$). errors (linear $z = 0.39$, $p = .700$), or standardized performance costs (linear $t = 0.42$, $p = .672$). As can be seen in the top row of Fig. 3, matching effects were present from the beginning. This pattern of results is most compatible with the assumption that the cognitive system encodes sequences in a relational manner from the outset.

## 3. Experiment 2: Different-element chunks

In the next experiment, we wanted to generalize the finding from Experiment 1 that matching chunk grammars benefit performance, in a critical manner. Experiment 1 had shown strong effects of abstract inter-chunk relationships with sequential elements that had no natural ordering. However, the situation in that experiment was still fairly limited in that, just in the earlier work on this topic (Collard & Povel, 1982; Restle, 1970), shared sets of sequential elements were used across chunks. The use of shared elements can make abstract relationships

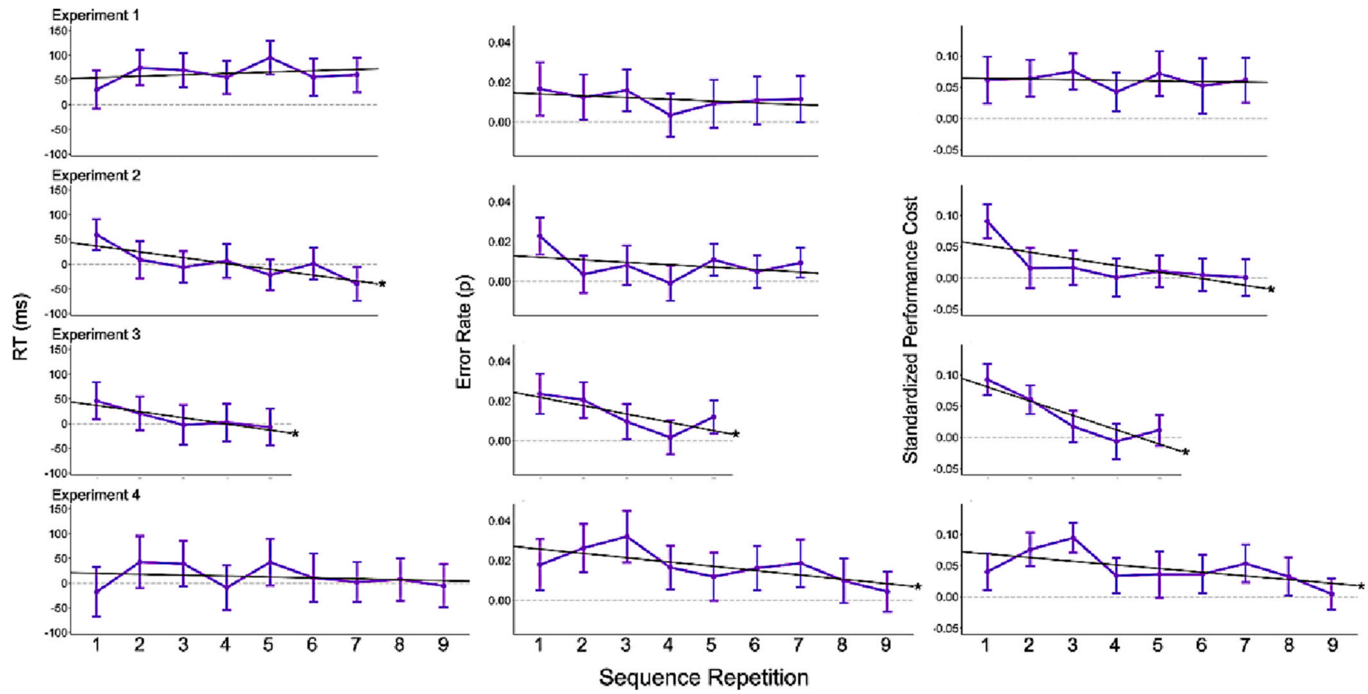


**Fig. 3.** Difference scores in RTs (left panels), error rates (middle panels), and standardized performance costs (right panels) for matching versus non-matching sequences across repetitions in all experiments (calculated as non-matching RT – matching RT). The gray dashed line marks zero (no difference between matching and non-matching). The regression line is shown in black, with an asterisk to the right indicating significance ($p < .05$). Error bars represent 95% within-subject confidence intervals.

highly salient. More importantly, when inter-chunk relationships are established across shared elements, the transitioning between chunks is often particularly straightforward. For example, Restle (1970) in his work used sequences such as 5–6-5—3-4-3—1–2–1. Here, the transition between chunks is almost trivial, once it is recognized that the "-2" operation transforms chunk $n$ into chunk $n + 1$. Similarly, in our Experiment 1, a sequence such as A-B-B—B-A-A allows a simple chunk-to-chunk transformation through an "inversion" operation (i.e., switching A to B and vice versa). Consistent with the results of Experiment 1, such between-chunk transformations should have particularly strong effects during chunk transitions because they put each chunk element in its place without requiring additional element-by-element retrieval. It is therefore important to examine the degree to which matching-grammar benefits occur even when elements across chunks are not shared and such straightforward between-chunk transformations are not possible. In Experiment 2, each sequence contained one chunk of horizontal/vertical rules (like in Experiment 1) and one chunk of clockwise/counterclockwise rules. Thus, in this experiment, abstraction across the identity of sets of elements was required in order to capitalize on the matching chunk patterns.

### 3.1. Methods

Given the robust effects in the Experiment 1, for this version we collected data from 40 participants who completed 36 blocks of 48 trials (36 six-element sequences, each repeated 8 times) in a 1.5-h session. Two participants did not complete the study and were therefore excluded, and the data from 38 participants were used in analyses. Exclusion criteria and analysis plan were the same as in Experiment 1.

### 3.2. Results

Fig. 2 (column 2) shows the results of this experiment. While there was no effect of shared chunk pattern on RTs and only a marginal effect on standardized performance scores, there was a small but highly robust effect on error rates (Table 2). Though there was a slight, non-significant tendency for matching benefits at chunk transitions for RTs, there was a counteracting effect specific to chunk transitions for errors — a pattern that may indicate a speed-accuracy tradeoff, given that this effect was markedly non-significant for standardized performance costs. No matching effects on the sequence transition level were observed (i.e., the ST contrast). Overall, the matching effects were much more subtle than in Experiment 1, consistent with the interpretation that in the first experiment, shared elements may have facilitated the detection and effective utilization of abstract relationships.

As in Experiment 1, we again examined how the matching effects developed across sequence repetitions. Interestingly, the pattern of repetition effects was, if anything, counter to a gradual learning hypothesis. For both RTs ($t = 3.56$, $p < .001$) and performance costs ($t = 3.38$, $p = .001$), there was a statistically robust linear decrease of matching benefits across repetitions; for errors there was no reliable linear trend ($z = 0.39$, $p = .699$). We will discuss the unexpected pattern of diminishing matching effects in concert of the results in the remaining experiments in the General Discussion (Section 6). However, we can conclude that again, results are clearly not consistent with a gradual learning perspective.

## 4. Experiment 3: Different-element chunks with auditory instructions

Experiment 2 provided evidence for small, but robust matching benefits. However, a potential concern with the procedure of the first two experiments is that the visual presentation of each rule sequence prior to the block may have facilitated the detection of abstract patterns among the simultaneously visible, individual elements in the instruction screen. Therefore, in Experiment 3, we checked whether we could replicate the matching effect when sequence positions were presented in a serial, auditory manner.

**Table 2**
Fixed effects from linear mixed models for Experiment 2.

| | RT | | | | Error Rate | | | | Standardized Performance Score | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | B | SE | t | p | B | SE | z | p | B | SE | t | p |
| Match | −0.001 | 0.009 | −0.14 | 0.889 | −0.234 | 0.066 | −3.54 | <0.001 | −0.020 | 0.011 | −1.89 | 0.060 |
| CT | 0.227 | 0.003 | 82.11 | <0.001 | 0.032 | 0.028 | 1.15 | 0.250 | 0.208 | 0.007 | 27.95 | <0.001 |
| WC | 0.008 | 0.003 | 2.50 | 0.012 | 0.192 | 0.033 | 5.78 | <0.001 | 0.026 | 0.009 | 2.98 | 0.003 |
| ST | 0.043 | 0.005 | 9.50 | <0.001 | −0.125 | 0.045 | −2.75 | 0.006 | 0.026 | 0.012 | 2.17 | 0.030 |
| Match * CT | −0.008 | 0.005 | −1.78 | 0.075 | 0.095 | 0.051 | 1.86 | 0.063 | 0.000 | 0.011 | 0.01 | 0.992 |
| Match * WC | −0.009 | 0.005 | −1.64 | 0.100 | −0.098 | 0.063 | −1.57 | 0.116 | −0.020 | 0.012 | −1.65 | 0.099 |
| Match * ST | 0.002 | 0.008 | 0.27 | 0.784 | 0.023 | 0.081 | 0.28 | 0.780 | 0.004 | 0.017 | 0.26 | 0.794 |
| Model Stats | AIC | $cR^2$ | $mR^2$ | ICC | AIC | $cR^2$ | $mR^2$ | ICC | AIC | $cR^2$ | $mR^2$ | ICC |
| | 74,076 | 0.235 | 0.146 | 0.104 | 17,361 | 0.083 | 0.012 | 0.071 | −557 | 0.85 | 0.524 | 0.685 |

*Note.* For RT and Error Rates, the following model was used: Outcome ~ Match * Position Contrasts + (1 + Match|Subject). Due to convergence issues, a simplified model was used for standardized performance score: Outcome ~ Match * Position Contrasts + (1|Subject).Match = matching grammars within sequence. CT = chunk transition contrast: first and fourth elements vs. all others (second, third, fifth, and sixth). WC = within-chunk contrast: second and fifth elements vs. third and sixth. ST = sequence transition contrast: first vs. fourth element. AIC = Akaike information criterion, $cR^2$ = conditional $R^2$, $mR^2$ = marginal $R^2$, ICC = intraclass correlation coefficient.

### 4.1. Methods

We collected data from 45 participants. Four participants did not meet the trial-wise inclusion criteria and were therefore excluded, leaving 41 participants. Exclusion criteria and analysis plan were the same as in Experiments 1–2.

Participants completed 36 blocks of 36 trials (36 six-element sequences, each repeated 8 times) in a 1.5-h session. The same sequence construction was used here as in Experiment 2. However, the instruction and error screens were different. Instead of displaying a sequence of rules on the screen at the beginning of each block, participants were shown a screen indicating that they should listen to the sequence of rules in their headphones. The six rules were presented one at a time for 1 s each, using a female text-to-speech voice. Participants could replay the sequence as many times as they wanted before pressing a key to begin the block.

Like in the previous experiments, after an incorrect response, the current rule would be displayed in red above the frame, in its position within the sequence. However, the rest of the rules in the sequence were represented by white dashes to indicate the position without allowing multiple rules in the sequence to be presented visually at the same time. Participants were then required to correct their mistake before moving on to the next trial. Because the auditory presentation format took extra time, we reduced the number of sequence cycles per block from 8 to 6 repetitions in order to fit the entire procedure within the experimental session.

### 4.2. Results

As shown in Fig. 2 (column 3), results of this experiment revealed small and robust effects of matching benefits overall for error rates and standardized performance cost scores, but not for RTs. In addition, we found robust RT matching benefits at chunk transitions, which was somewhat offset by counteracting error effects, and therefore yielded no reliable effect for standardized performance cost (Table 3). Again, no matching effects on the level of sequence transitions were observed.

Experiment 3 also replicated the unexpected interaction between cycle repetitions and matching benefits found in Experiment 2 (Fig. 3, third row). Here, for RTs ($t = 4.16$, $p < .001$), error rates ($z = 2.21$, $p = .027$), and performance costs ($t = 4.07$, $p < .001$), the effect of matching chunk patterns was strongest for earlier rounds of sequence execution and then dissipated linearly.

## 5. Experiment 4: Second-order matches

In this final experiment, we wanted to test the limits of the small, but robust abstract matching effect we had obtained in the previous two experiments with non-overlapping chunks. Specifically, we examined whether or not participants are able to benefit from second-order relationships. For this purpose, we created 12-element sequences constructed from two six-element subsequences containing two three-element chunks each. We will refer to these subsequences as "chunk plans". The three-element chunks within each chunk plan used the same element pairs, as in Experiment 1. However, there were no shared elements across the two chunk plans. The critical match here was between the structures of the two chunk plans. Matching sequences had plans with similar chunk structures (e.g., A-B-A—B-B-A——C-D-C—D-D-C); non-matching sequences had plans with dissimilar chunk structures (e.g., A-B-A—B-B-A——C-D-C—D-C-C. Importantly, chunk plans were constructed from pairs of three-element chunks with no direct chunk

**Table 3**
Fixed effects from linear mixed models for Experiment 3.

| | RT | | | | Error Rate | | | | Standardized Performance Score | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | B | SE | t | p | B | SE | z | p | B | SE | t | p |
| Match | −0.009 | 0.012 | −0.73 | 0.468 | −0.223 | 0.096 | −2.31 | 0.021 | −0.036 | 0.015 | −2.45 | 0.018 |
| CT | 0.254 | 0.003 | 79.73 | <0.001 | 0.028 | 0.030 | 0.95 | 0.340 | 0.225 | 0.009 | 25.98 | <0.001 |
| WC | 0.014 | 0.004 | 3.94 | <0.001 | 0.128 | 0.035 | 3.68 | <0.001 | 0.026 | 0.010 | 2.59 | 0.010 |
| ST | 0.022 | 0.005 | 4.24 | <0.001 | −0.156 | 0.048 | −3.23 | 0.001 | 0.003 | 0.014 | 0.22 | 0.829 |
| Match * CT | −0.014 | 0.005 | −2.62 | 0.009 | 0.115 | 0.055 | 2.10 | 0.035 | −0.003 | 0.012 | −0.27 | 0.788 |
| Match * WC | −0.007 | 0.006 | −1.17 | 0.241 | −0.070 | 0.067 | −1.04 | 0.300 | −0.015 | 0.014 | −1.08 | 0.280 |
| Match * ST | −0.004 | 0.009 | −0.43 | 0.666 | 0.056 | 0.087 | 0.65 | 0.518 | 0.004 | 0.020 | 0.22 | 0.823 |
| Model Stats | AIC | $cR^2$ | $mR^2$ | ICC | AIC | $cR^2$ | $mR^2$ | ICC | AIC | $cR^2$ | $mR^2$ | ICC |
| | 56,442 | 0.295 | 0.164 | 0.157 | 14,545 | 0.163 | 0.009 | 0.155 | −395 | 0.848 | 0.416 | 0.740 |

*Note.* For each of the outcomes, the following model was used: Outcome ~ Match * Position Contrasts + (1 + Match|Subject). Match = matching grammars within sequence. CT = chunk transition contrast: first and fourth elements vs. all others (second, third, fifth, and sixth). WC = within-chunk contrast: second and fifth elements vs. third and sixth. ST = sequence transition contrast: first vs. fourth element. AIC = Akaike information criterion, $cR^2$ = conditional $R^2$, $mR^2$ = marginal $R^2$, ICC = intraclass correlation coefficient.

pattern matches. Therefore, pattern matches on the cross-plan level were not confounded with any potential matches on the within-plan level (which were the focus of the preceding experiments).

### 5.1. Methods

Participants ($n = 44$) completed 12 blocks of 120 trials (12 long sequences, each repeated 10 times) in a 1-h session. Four participants did not complete the study and were therefore excluded, and the data from 40 participants were used in analyses. Exclusion criteria were the same as in Experiments 1–3, and analysis plan was similar, with a few necessary differences, described below.

### 5.2. Results

To reflect the more complex, hierarchical structure of the sequences used in this experiment, in addition to the CT contrast (positions 1, 4, 7, and 10, vs. rest), we used a plan transition contrast (PT) comparing positions 1 and 7 vs. 4 and 10, and a sequence-level transition contrast (ST) testing position 1 against position 7.

Again, Fig. 2 (column 4) and Table 4 show the results. In all three performance outcomes, we found clear evidence for matching benefits. For RTs, these effects were slightly larger at plan and chunk transition points. On the level of sequence transitions, there seemed to be opposing tendencies between RTs and errors, again potentially reflecting a speed-accuracy tradeoff, given the non-significant effect of performance costs. Overall, these results indicate that even high-level abstract relationships (i.e., relationships between relationships) are appreciated during sequence production. Additionally, we again replicated the unexpected pattern of dissipating matching benefits as a linear function of sequence repetitions, but only for error scores ($z = 2.40$, $p = .016$) and performance costs ($t = 2.21$, $p = .027$), and not for RTs ($t = 0.48$, $p = .631$).

## 6. The role of element repetitions

The sequences we used were constructed from chunks in which elements (i.e., rules) either repeated or changed across consecutive trials. Because any chunk could occur both in matching or non-matching sequences, this aspect was not confounded with the critical grammar-matching contrast. However, rule changes have been shown to produce substantial effects on performance, akin to task-switching costs (e.g., Mayr, 2002). Also, in the absence of any other natural ordering between elements, sequence grammars can only be established through the pattern of element repetitions. Therefore, it is prudent to examine whether and how repetitions/changes of elements modulate the matching effect in an exploratory manner. Not surprisingly, repetitions were generally performed faster and more accurately than changes (Exp.1: RT: $t = 48.04$, $p < .001$; Error Rate: $z = 2.57$, $p = .010$; Performance Cost: $t = 17.49$, $p < .001$; Exp. 2: RT: $t = 33.08$, $p < .001$; Error Rate: $z = 2.41$, $p = .016$; Performance Cost: $t = 13.45$, $p < .001$; Exp. 3: RT: $t = 23.22$, $p < .001$; Error Rate: $z = -3.47$, $p = .001$; Performance Cost: $t = 7.08$, $p < .001$; Exp. 4: RT: $t = 32.00$, $p < .001$; Error Rate: $z = 0.31$, $p = .757$; Performance Cost: $t = 10.92$, $p < .001$). Fig. 4 shows the average matching contrast for each experiment as a function of repetitions and changes. This analysis is complicated by the fact that for Experiments 2–4, we used non-shared sets of rules for the first versus second half of each sequence. This constrains the first positions of each half (i.e., each chunk in Exp. 2 and 3, each chunk plan in Exp. 4) to exclusively contain rule changes. Therefore, Fig. 4 shows results for change trials in two ways: using only those positions that are directly comparable with the repetition trials (solid red lines), and including all change trials (dashed red lines).

While in Experiment 1 the change/repeat contrast does not affect the matching benefit, in Experiments 2–4 this effect was clearly stronger for rule repetitions than for rule changes. For rule changes, error benefits remained reliable, albeit in a reduced manner, whereas in RTs matching grammars appeared to produce costs. Thus, while the matching contrast has effects in all conditions, it is only in the repeat trials that it leads to clear benefits. In change trials, results point to a shift in the speed-accuracy tradeoff criterion. Irrespective of these speed-accuracy complexities for change trials, the pattern of considerably larger matching benefits for repeat trials is an important, albeit unexpected, result. It suggests that repetitions are not just critical for the detection of abstract relationships, but also for their utilization during sequence production. We will return to this aspect in the General Discussion (Section 6.4).

## 7. General discussion

Across four experiments, we tested whether abstract chunk relationships are registered and used by the cognitive system. Past research has provided evidence for the utilization of abstract patterns, but was mostly limited to situations in which sequential elements had inherent ordering properties that might support the detection of patterns and where detection of patterns could be accomplished through gradual, feedback-based hypothesis testing and learning. We were therefore interested in determining the degree to which utilization of abstract sequential patterns is a general property of sequential representations. An affirmative answer to this question would have important theoretical implications for current models of hierarchical control. Such models typically assume that the hierarchical organization of sequential representations supports a divide-and-conquer strategy, in which different chunks of a sequence assume isolated representational subspaces in order to avoid between-chunk interference. Therefore, it is not clear how such models could at the same time account for the across-chunk integration that would be necessary to utilize abstract relationships.

**Table 4**
Fixed effects from linear mixed models for Experiment 4.

| | RT | | | | Error Rate | | | | Standardized Performance Score | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | B | SE | t | p | B | SE | z | p | B | SE | t | p |
| Match | −0.026 | 0.020 | −1.28 | 0.208 | −0.378 | 0.091 | −4.17 | <0.001 | −0.064 | 0.022 | −2.89 | 0.005 |
| CT | 0.089 | 0.002 | 45.54 | <0.001 | −0.087 | 0.017 | −5.12 | <0.001 | 0.066 | 0.004 | 14.61 | <0.001 |
| PT | 0.137 | 0.004 | 33.52 | <0.001 | −0.091 | 0.036 | −2.53 | 0.012 | 0.105 | 0.009 | 11.12 | <0.001 |
| ST | 0.026 | 0.009 | 2.96 | 0.003 | −0.053 | 0.081 | −0.65 | 0.516 | 0.015 | 0.020 | 0.77 | 0.442 |
| Match * CT | −0.011 | 0.003 | −4.18 | <0.001 | −0.017 | 0.027 | −0.65 | 0.514 | −0.010 | 0.006 | −1.53 | 0.125 |
| Match * PT | −0.013 | 0.006 | −2.20 | 0.028 | −0.042 | 0.056 | −0.74 | 0.456 | −0.011 | 0.013 | −0.85 | 0.394 |
| Match * ST | −0.019 | 0.012 | −1.52 | 0.127 | 0.253 | 0.129 | 1.97 | 0.049 | 0.003 | 0.028 | 0.09 | 0.926 |
| Model Stats | AIC | $cR^2$ | $mR^2$ | ICC | AIC | $cR^2$ | $mR^2$ | ICC | AIC | $cR^2$ | $mR^2$ | ICC |
| | 74,754 | 0.211 | 0.079 | 0.144 | 17,567 | 0.154 | 0.017 | 0.140 | −324 | 0.677 | 0.165 | 0.613 |

*Note*. For each of the outcomes, the following model was used: Outcome ~ Match * Position Contrasts + (1 + Match|Subject). Match = matching grammars within sequence. CT = chunk transition contrast: first, fourth, seventh, and tenth elements vs. all others. PT = chunk plan transition contrast: first and seventh elements vs. fourth and tenth. ST = sequence transition contrast: first vs. seventh element. AIC = Akaike information criterion, $cR^2$ = conditional $R^2$, $mR^2$ = marginal $R^2$, ICC = intraclass correlation coefficient.

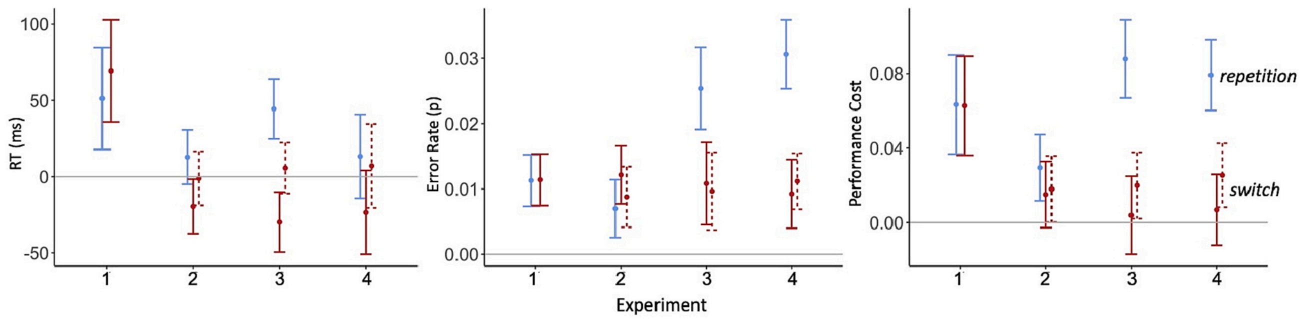


**Fig. 4.** Average difference scores (nonmatch - match) for each experiment as a function of element repetitions versus switches, for all three outcomes. Error bars reflect the 95% within-subject confidence interval. Solid error bar lines were calculated only from sequence positions for which both switches and repetitions were possible (e.g., chunk positions 2–3 in Experiment 2), dashed error bar lines from element switches at all sequence positions, including those for which element repetitions were not possible (chunk positions 1 in Experiments 2 and 3, chunk-plan position 1 in Experiment 4).

### 7.1. Utilization of abstract relationships

Answering our primary research question, we found across four different experiments robust evidence that abstract relationships between different parts of a sequence do indeed lead to performance benefits. Importantly, in our paradigm the basic elements were response rules (e.g., horizontal vs. vertical). Thus, these results confirm that the utilization of abstract relationships can be generalized beyond stimuli with inherent ordering properties, such as numbers or spatial locations. In the absence of inherent ordering principles, a grammar that organizes sequential elements can only express itself through the pattern of element repetitions (e.g., AAB vs. ABA vs. ABB). In fact, it has been claimed that the human system is uniquely equipped for detecting and utilizing patterns of element repetitions (Cho et al., 2002; Dehaene et al., 2015, 2022). It is also noteworthy that participants were able to utilize even second-order abstract relationships, that is when the relationships between two chunks in the first half of the sequence matched the relationship between the two chunks in the second half of the sequence (despite different elements across the two chunk plans in the larger sequence).

### 7.2. The role of sequence positions

For our second research question, we wanted to clarify if abstract relationships affect all sequential elements, or instead are particularly expressed during chunk transitions (e.g., at the beginning of a chunk or chunk plan). For all four experiments, we found robust error benefits that extended across all sequence elements. However, Fig. 2 also clearly indicates that the results of Experiment 1 stand out from those of the remaining experiments. That experiment used shared elements across chunks and showed a very strong RT benefit for matching chunks at transition points. While interactions between the match factor and transitions contrasts were also found in the remaining experiments, these were much more subtle and in part offset by speed-accuracy effects.

What may explain the particularly large transition benefits only in Experiment 1? We suggest that for matching chunks with shared elements, the abstract relationship can be used to directly translate the preceding chunk in the following chunk through an "inversion" operation, where every element in chunk 1 is turned into the alternative possible element in chunk 2. This eliminates any need for retrieving individual elements and placing them into their "slots," as is necessary when elements are not shared. In that sense, the result of Experiment 1 is very similar to the body of results from earlier work with abstract relationships in sequences with ordered elements (Restle, 1970; Restle & Burnside, 1972). Here too, the abstract relationships could be used to directly transform one part of a sequence into the next through the various possible transformation rules, leading to strongest benefits at transition points. However, beyond this direct translation effect during transitions, our results consistently show a second, more subtle matching benefit that affects all sequential elements equally. Importantly, this implies that abstract relationships between chunks, allow a more efficient representation of the entire sequence, not only at chunk transitions.

### 7.3. The role of within-sequence experience

The third question we wanted to address is particularly important. There is previous evidence that people (and even infants) are able to capitalize on abstract sequential patterns (Aslin & Newport, 2012; Marcus et al., 1999). These results stem from learning paradigms that allowed the gradual discovery of "hidden" relationships (see also, Collins & Frank, 2013). However, it is also possible that our cognitive system is prepared towards encoding sequential material from the outset in terms of relationships between elements and sequence parts. If that is the case, we should see abstract pattern benefits emerging from the beginning, rather than gradually as a function of repeated exposure with the same sequence. Our results were very clear in this regard. We found that abstract benefits were present from the beginning in all of our experiments. Further, in each of Experiments 2–4, we found for at least one of the two dependent variables the opposite of what would be predicted by gradual learning, namely a decrease of the matching benefit (see Fig. 3). Indeed, for Experiment 2 and 3 we also found a reduction of an initial RT benefit—a result that also makes the finding of overall matching effects only in errors and not in RTs a bit less surprising.

While unexpected, the "reverse" learning effect is consistent with an "expectation of relationships" hypothesis. To see why, it is important to appreciate that even our non-matching sequences could allow inter-chunk relationships to be established. For example, a sequence such as ABB-DDC can be described as a mirroring of the initial abstract order. Thus, while our matched sequences allowed participants to pick up the relationships from the outset, the cognitive system might need more experience (i.e., a few cycles) with each new sequence to detect and utilize the more subtle relationships present in our "non-matching" conditions.

This perspective also shines light on another aspect of our results. In Experiments 2–4, our matching effects were subtle, and one might wonder how seriously one should take such small effects as a marker of an important principle of sequential representations. However, if the cognitive system is geared towards constructing relationships even when they are not obvious, it should be difficult or impossible to create an all-or-none contrast between sequences that contain abstract relationships and those that do not. The actual contrast we can establish will be much more graded and therefore subtle effects should not come as a surprise.

### 7.4. Theoretical implications

In this final section, we turn to the question how our results fit with current models of hierarchical, serial-order control. One long-standing question in this context is to what degree serial order is established through some form of associative chaining between specific elements versus through abstract, element-independent position codes. There is evidence that through statistical learning of successive element relationships, abstract patterns can be extracted. However, such learning occurs gradually, across a number of repetitions (Botvinick & Plaut, 2004; Davachi & DuBrow, 2015). Therefore, the finding that abstract patterns for new sequences can be extracted and used from the outset clearly favors the abstract-position account (see also, Kikumoto & Mayr, 2018; Mayr, 2009).

The second, broader question is how our results that indicate integration of information across chunk boundaries can be reconciled with the idea that hierarchical control implements a divide and conquer approach towards resolving interference across different regions of a complex task space (Badre, 2008; Badre & Nee, 2018; Koechlin et al., 2003; Koechlin & Jubault, 2006). In principle, the hierarchical structure of a memorized sequence can be viewed as an LTM retrieval structure (e.g., Ericsson & Kintsch, 1995), where relationships between chunks are part of declarative memory and are loaded into WM during chunk transitions as cues to retrieve the next chunk. As discussed in section 6.2, this can explain the large benefits of matching grammars on the first position of a chunk in Experiment 1, where shared elements across related chunks allowed the straightforward translation of one chunk into the next during transitions. However, additional considerations are required to explain the fairly uniform effect of matching grammars across chunk positions in the remaining experiments.

We believe that this uniform matching effect is related to another recent finding from our lab (Moss & Mayr, 2023). Specifically, we used similar task sequences as in the current experiments but manipulated the number of relevant hierarchical levels between 1 and 4 levels in a blockwise manner. We found that RTs and error rates scaled with the number of levels, even for within-chunk elements, that is when no chunk transition was required. In other words, operations within a given chunk were not protected from the overall dimensionality of the task space. Interestingly, in hierarchical structures that were cued from the environment, rather than based on sequential plans, no comparable number-of-levels effects were found for non-transition points. We interpreted this overall pattern in terms of a unique requirement within sequential representations, namely of keeping track of the current position within the overall sequence's state space—something that is not needed for cue-based hierarchical structures. Importantly, the dimensions of the state space of a sequence are specified by the hierarchical levels. Therefore, every trial, not just chunk transitions, should be the more demanding as more hierarchal levels need to be considered in order to keep track of the current position—thus giving rise to the uniform number-of-levels effect.

If it is correct that every trial of a sequence poses the problem of localizing oneself within the larger sequence, then this also provides opportunity for inter-chunk relationships to be helpful on every trial. How might this happen? The long-term WM (LT-WM) model proposed by Ericsson and Kintsch (1995) provides a possible answer. This model was originally devised to explain exceptional performance in serial-memory tasks through the use of hierarchical, LTM retrieval structures that expand traditional short-term memory capacity limitations. An important obstacle for high performance in serial-memory tasks is proactive interference across chunks (e.g., from the element associated to position 2 of chunk $n$-1 to the element associated with position 2 in chunk $n$; Henson, 1996; Mayr, 2009). Importantly, a key assumption of the LT-WM model is that encoding individual sequential elements in terms of relational schemes, instead of as arbitrary chunk and/or position codes, protects against proactive interference (Bjork, 1978). From this perspective, relational coding within and across chunks would provide a solution to the very interference problem that the separation into distinct subspaces through hierarchical control structures is meant to address. Importantly, relational coding reduces interference for every sequential element, leading to the uniform benefits of matching grammars that we have observed.

Another important aspect of the LT-WM model is that it assumes that declarative knowledge is used for building relationships between elements. This fits well with the observation that abstract relationships were utilized from the outset. Most likely, this occurs already during initial encoding of the sequences, allowing the integration of inter-chunk relationships into the representation of each sequence. Thus, while there is evidence suggesting that hierarchical structures and relationships can be acquired in an implicit and gradual manner (e.g., Collins & Frank, 2013; Marcus et al., 1999), the current results suggest that the cognitive system has a basic preparedness to code each piece of new information in relation to preceding information (Cho et al., 2002; Dehaene et al., 2022).

Finally, we found that immediate repetitions/changes strongly modulated the grammar-matching effect. Overall, matching benefits were highly robust only for element (i.e., rule) repetitions across consecutive trials. For change trials, matching grammars still influenced performance, but in a manner that was most consistent with a speed-accuracy tradeoff. Further, this pattern was only present for Experiments 2–4, which used non-shared elements. For Experiment 1, matching-grammar effects were the same for change and repeat trials, reinforcing the interpretation that relationships are utilized in different ways when chunks share elements than when they do not. While the modulation of the matching-grammar effect through the change/repetition factor was unexpected, it is sufficiently consistent to justify speculation about the mechanisms behind it. One potentially important aspect is that optimal performance in task-switching situations, such as in our rule-selection task, requires working memory to adequately toggle between a maintenance state on repeat trials, and an updating state on change trials (e.g., Mayr, Kuhns, & Hubbard, 2014; O'Reilly, 2006). Thus, it is possible that the cognitive system uses the inter-chunk relationships to optimize the maintenance/updating decision. More adequate maintenance through the use of relational information will have unmitigated benefits for performance. However, things may be more complicated for rule-change trials. Specifically, it has been shown that for predictable switches, response criteria increase, possibly to safeguard against premature responses based on a perseverated task set or rule (Schmitz & Voss, 2012). Thus, as relational coding of sequences makes switches more predictable, it will also lead to an increase of the response criterion and therefore to longer RTs.

Overall, our results are fully consistent with the view that the cognitive system comes with a built-in prior for expecting that sequences can be coded in terms of relationships between successive elements and chunks. This conclusion is very similar to those by Restle (1970) and others, but extends it to sequences without ordered elements and to a paradigm without feedback-driven learning and hypothesis testing. Recent models of hierarchical control have emphasized the question how our cognitive system is able to divide complex tasks into separable subspaces, rather than how information is integrated and shared across such subspaces. We do not believe that the current or other, similar results necessarily falsify "divide-and-conquer" models. However, they point to an overlooked category of observations that needs to be accommodated in any full model of hierarchical control.

## Credit author statement

M.E.M was involved in planning the studies, she programmed the experiments, helped run the experiments, helped analyze the data, and write the manuscript.

M.Z. was involved in planning the studies, helped run the experiment, and analyze the data.

U.M. oversaw all phases of the project including securing the funding. He was particularly involved in planning the study and writing of the manuscript.

### Data availability

Data and analyses are made available through OSF

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## Appendix A. Table of all possible sequences for each of the experiments

| | Match | Nonmatch | |
|---|---|---|---|
| Exp 1: | HHV – VVH | HHV – VHH | HHV – VHV |
| | VVH – HHV | VHH – HHV | VHV – HHV |
| | HVV – VHH | HVV – VVH | HVV – VHV |
| | VHH – HVV | VVH – HVV | VHV – HVV |
| | HVH – VHV | HVH – VVH | HVH – VHH |
| | VHV – HVH | VVH – HVH | VHH – HVH |
| Exp 2–3: | HHV – CCL \| CCL – HHV | HHV – CLL \| CLL – HHV | HHV – CLC \| CLC – HHV |
| | HHV – LLC \| LLC – HHV | HHV – LCC \| LCC – HHV | HHV – LCL \| LCL – HHV |
| | VVH – CCL \| CCL– VVH | VVH – CLL \| CLL – VVH | VVH – CLC \| CLC – VVH |
| | VVH – LLC \| LLC – VVH | VVH – LCC \| LCC – VVH | VVH – LCL \| LCL – VVH |
| | HVV – CLL \| CLL – HVV | HVV – CCL \| CCL – HVV | HVV – CLC \| CLC – HVV |
| | HVV – LCC \| LCC – HVV | HVV – LLC \| LLC – HVV | HVV – LCL \| LCL – HVV |
| | VHH – CLL \| CLL – VHH | VHH – CCL \| CCL – VHH | VHH – CLC \| CLC – VHH |
| | VHH – LCC \| LCC – VHH | VHH – LLC \| LLC – VHH | VHH – LCL \| LCL – VHH |
| | HVH – CLC \| CLC – HVH | HVH – CCL \| CCL – HVH | HVH – CLL \| CLL – HVH |
| | HVH – LCL \| LCL – HVH | HVH – LLC \| LLC – HVH | HVH – LCC \| LCC – HVH |
| | VHV – CLC \| CLC – VHV | VHV – CCL \| CCL – VHV | VHV – CLL \| CLL – VHV |
| | VHV – LCL \| LCL – VHV | VHV – LLC \| LLC – VHV | VHV – LCC \| LCC – VHV |
| Exp 4: | HHV–VHH — LLC–CLL \| LLC–CLL — HHV–VHH | HHV–VHH — LCC–CLC \| LCC–CLC — HHV–VHH | |
| | HHV–VHV — LLC–CLC \| LLC–CLC — HHV–VHV | HHV–VHV — LCC–CLC \| LCC–CLC — HHV–VHV | |
| | HVV–VHV — LCC–CLC \| LCC–CLC — HVV–VHV | HVV–VVH — LCC–CLC \| LCC–CLC — HVV–VVH | |
| | HVV–VVH — LCC–CCL \| LCC–CCL — HVV–VVH | HVV–VHV — LCC–CCL \| LCC–CCL — HVV–VHV | |
| | HVH–VHH — LCL–CLL \| LCL–CLL — HVH–VHH | HVH–VVH — LCL–CLL \| LCL–CLL — HVH–VVH | |
| | HVH–VVH — LCL–CCL \| LCL–CCL — HVH–VVH | HVH–VVH — LCC–CLC \| LCC–CLC — HVH–VVH | |

*Note.* H = horizontal, V = vertical, L = clockwise, C = counterclockwise. '|' indicates that participants were randomly given one of the two sequences to execute ('or').

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